Metric Emergence from Event-Order Geometry

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Abstract

Papers XXIII–XXIV derived a strict partial order on admissible overlap events and showed that observer-dependent time functions, relativity of simultaneity, and time-dilation analogues emerge from the event poset. The present paper addresses the next question: what does it mean for two events to be *near or far*?

Three candidate separation constructions are developed and compared. For comparable events (those related by the partial order), a *chain-length separation* counts the minimum number of cover steps between them, providing an observer-independent temporal interval. For incomparable events (those not causally related), an *observer-disagreement separation* measures the variance of time-difference assignments across observers, providing a structural precursor to spatial distance. A *transition-cost separation* on the Hasse graph gives a unified graph-distance that applies to both cases.

The chain-length separation is proved to be a well-defined directed temporal separation on comparable pairs. The observer-disagreement construction is shown to exhibit observer-independent sign for comparable pairs and observer-dependent sign for incomparable pairs, with the variance quantity providing a nontrivial separation measure. When the transition graph is connected, the transition-cost construction satisfies all metric axioms on the event set. A worked example on the dual-pentagon system demonstrates all three constructions explicitly.

The paper does not derive the Lorentz metric or continuous geometry. It establishes that the event-order framework supports a nontrivial separation structure, providing a concrete route from poset to emergent metric geometry.

1 Introduction

Paper XXIII [Smart, 2026a] derived a strict partial order $\prec$ on an event set $\mathcal{E}$ from phase-overlap geometry. Paper XXIV [Smart, 2026b] showed that observers (linear extensions with sampling measures) produce frame-dependent time functions, relativity of simultaneity, and time-dilation analogues.

These results give the event set temporal and observer structure, but not spatial or metric structure. The present paper asks: *given only the event poset and its observer completions, can one define a meaningful notion of separation between events?*

The essential distinction

Two events can be related in exactly two ways under $\prec$:

- **Comparable** ($e_1 \prec e_2$ or $e_2 \prec e_1$): causally / temporally ordered.
- **Incomparable** (neither holds): "simultaneous" / spacelike-separated.

Any separation construction must handle both cases, and the two cases may require different structures. The paper develops three candidates and compares them.

Scope

This paper constructs separation quantities and proves their basic properties. It does not derive the Lorentz metric, continuous manifold structure, or the Einstein equations. It establishes that the event-order framework naturally supports emergent metric-like geometry.

2 Cover Relations and the Transition Graph

Definition 1 (Cover relation)**.** *Event e_1 is covered by e_2 (written $e_1 \lessdot e_2$) if $e_1 \prec e_2$ and there is no event e with $e_1 \prec e \prec e_2$.*

Definition 2 (Transition graph (Hasse graph))**.** *The transition graph $\mathcal{G} = (\mathcal{E}, E_H)$ has vertex set $\mathcal{E}$ and edge set*

$$E_H = \{(e_1, e_2) : e_1 \lessdot e_2\} \tag{1}$$

This is the Hasse diagram of $(\mathcal{E}, \prec)$, viewed as an undirected graph (ignoring edge orientation for the purpose of graph distance).

Remark 3. *The cover relation encodes "immediate succession": $e_1 \lessdot e_2$ means e_2 is the next event after e_1 with no intervening event in the partial order. The transition graph captures the minimal-step accessibility structure of the event set.*

3 Construction A: Chain-Length Separation

For comparable events, the natural separation is the number of order steps between them.

Definition 4 (Chain-length separation)**.** *For comparable events $e_1 \prec e_2$, define:*

$$d_{\text{chain}}(e_1, e_2) = \min\{|C| - 1 : C \text{ is a chain from } e_1 \text{ to } e_2\} \tag{2}$$

where $|C|$ is the number of events in C. For $e_1 = e_2$, set $d_{\text{chain}}(e, e) = 0$.

Remark 5. *$d_{\text{chain}}(e_1, e_2) = k$ means there is a chain $e_1 = c_0 \prec c_1 \prec \cdots \prec c_k = e_2$ of length k, and no shorter chain exists. This is the minimum number of "causal steps" between e_1 and e_2.*

Proposition 6 (Properties of chain-length separation)**.** *On the set of comparable pairs, d_{chain} satisfies:*

1. ***Non-negativity:*** $d_{\text{chain}}(e_1, e_2) \geq 0$.

2. ***Identity:*** $d_{\text{chain}}(e, e) = 0$.

3. ***Triangle inequality:*** *If* $e_1 \prec e_2 \prec e_3$*:*

$$d_{\text{chain}}(e_1, e_3) \leq d_{\text{chain}}(e_1, e_2) + d_{\text{chain}}(e_2, e_3) \tag{3}$$

Proof. (1): Chain lengths are non-negative integers. (2): The trivial chain $\{e\}$ has $|C| - 1 = 0$. (3): Concatenating a shortest chain from e_1 to e_2 with one from e_2 to e_3 gives a chain from e_1 to e_3 of length $d_{\text{chain}}(e_1, e_2) + d_{\text{chain}}(e_2, e_3)$. The minimum over all chains can only be shorter. □

Remark 7. *d_{chain} is a directed separation function, not a metric: it is only defined for comparable pairs and is inherently directional (from earlier to later). Incomparable events have no chain between them, so d_{chain} does not apply. This separation is temporal: it measures causal distance, not spatial distance.*

Remark 8 (Longest chain as proper-time analogue)**.** *In causal set theory, the longest (not shortest) chain between comparable events is the analogue of proper time. Define:*

$$d_{\max}(e_1, e_2) = \max\{|C| - 1 : C \text{ is a chain from } e_1 \text{ to } e_2\} \tag{4}$$

$d_{\max}$ is well-defined for comparable pairs but does not in general satisfy a metric-style triangle inequality. It is better viewed as an alternative temporal-size functional. In the VFD framework, whether d_{chain} (shortest) or $d_{\max}$ (longest) better approximates physical proper time is an open question; in causal set theory, the longest chain is preferred.

4 Construction B: Observer-Disagreement Separation

For incomparable events, no chain exists and d_{chain} is undefined. A different construction is needed.

Definition 9 (Observer-disagreement separation)**.** *For any two events $e_1, e_2 \in \mathcal{E}$, let $\mathcal{O}$ be a family of observers. Define:*

$$\sigma(e_1, e_2) = \mathrm{Var}_{O \in \mathcal{O}}[t_O(e_1) - t_O(e_2)] \tag{5}$$

the variance of the time-difference assignment across the observer family.

Proposition 10 (Observer-dependent sign structure)**.**

1. *If $e_1 = e_2$: $t_O(e_1) - t_O(e_2) = 0$ for all observers O.*

2. *If $e_1 \prec e_2$ (comparable): $t_O(e_1) - t_O(e_2) < 0$ for all observers O. The sign is observer-independent.*

3. *If e_1 and e_2 are incomparable: there exist observers O_1, O_2 such that*

$$t_{O_1}(e_1) - t_{O_1}(e_2) < 0, \qquad t_{O_2}(e_1) - t_{O_2}(e_2) > 0.$$

The sign is observer-dependent.

Consequently, the variance $\sigma(e_1, e_2)$ is zero for identical events, and generically nonzero for incomparable pairs (where the sign-changing contributions ensure positive variance).

Proof. (1): Immediate. (2): $e_1 \prec e_2$ implies $t_O(e_1) < t_O(e_2)$ for all O (Proposition 1 of Paper XXIV). (3): By the relativity-of-simultaneity theorem (Theorem 1 of Paper XXIV), linear extensions exist placing the pair in either order. □

Remark 11 (Interpretation)**.** *The observer-disagreement σ serves as a spatial-separation precursor: it is large for events that observers strongly disagree about, and zero only for identical events. Comparable events have constrained (sign-fixed) disagreement; incomparable events admit sign-varying disagreement across observers. This mirrors the physical intuition that timelike-separated events have a fixed ordering while spacelike-separated events do not.*

Remark 12 (Metric failure)**.** *σ does not satisfy the triangle inequality in general and is therefore not a metric. It is a meaningful separation measure but not a distance function in the strict sense. Its role is to provide nontrivial structure for incomparable pairs where d_{chain} is undefined.*

5 Construction C: Transition-Cost Metric

The transition graph $\mathcal{G}$ (Definition 2) provides a unified metric that applies to both comparable and incomparable events.

Definition 13 (Transition-cost metric)**.** *For any $e_1, e_2 \in \mathcal{E}$, define:*

$$d_{\text{trans}}(e_1, e_2) = \textit{shortest-path length in } \mathcal{G} \textit{ between } e_1 \textit{ and } e_2 \tag{6}$$

where the graph $\mathcal{G}$ is the undirected Hasse graph with unit edge weights.

Theorem 14 (Metric on $\mathcal{E}$)**.** *If $\mathcal{G}$ is connected, then d_{trans} is a metric on $\mathcal{E}$:*

1. *$d_{\text{trans}}(e_1, e_2) \geq 0$, with equality iff $e_1 = e_2$.*
2. *$d_{\text{trans}}(e_1, e_2) = d_{\text{trans}}(e_2, e_1)$.*
3. *$d_{\text{trans}}(e_1, e_3) \leq d_{\text{trans}}(e_1, e_2) + d_{\text{trans}}(e_2, e_3)$.*

Proof. These are the standard properties of shortest-path distance on a connected graph. (1): Shortest paths have non-negative length; length 0 iff $e_1 = e_2$. (2): Undirected graph, so paths are reversible. (3): Triangle inequality holds for shortest-path distance on any graph (concatenation gives an upper bound). □

Remark 15. *d_{trans} is the first genuine metric on the event set. Unlike d_{chain} (which requires comparability) and σ (which fails the triangle inequality), d_{trans} applies to all pairs and satisfies all metric axioms.*

Remark 16 (Comparable vs incomparable)**.** *For comparable events $e_1 \prec e_2$, the transition-cost distance satisfies $d_{\text{trans}}(e_1, e_2) \leq d_{\text{chain}}(e_1, e_2)$ (since any chain gives a path in the Hasse graph). In the pentagon toy model, equality holds: $d_{\text{trans}} = d_{\text{chain}} = 1$ for all comparable pairs. Whether equality holds in general depends on the specific Hasse-graph structure. For incomparable events, d_{trans} involves "sideways" steps through the graph — paths that go up then down (or vice versa) through the partial order. These sideways paths are the structural precursor to spatial separation.*

Table 1: Separation structure by event relationship.

Event pair	d_{chain}	σ	d_{trans}
Comparable	Defined (causal steps)	Sign-fixed	Graph distance bounded by cha
Incomparable	Undefined	Sign-varying (high variance)	Sideways path
Interpretation	Temporal interval	Spatial precursor	Unified metric

6 Temporal vs Spatial Separation

The three constructions cleanly separate into temporal and spatial-precursor components:

Remark 17 (Analogy to Lorentzian structure)**.** *In Lorentzian geometry, timelike intervals are computed along causal curves, and spacelike intervals require leaving the light cone. The VFD framework exhibits a structural analogue: comparable events have chain-based (causal-curve) separation, while incomparable events require sideways (off-chain) paths in the transition graph. The transition-cost metric d_{trans} does not distinguish timelike from spacelike — it is a single positive-definite distance. A signed interval $s^2 = d^2_{\text{chain}} - d^2_{\text{sideways}}$ could in principle be defined but is not pursued here.*

7 Observer Time vs Structural Interval

Proposition 18 (Structural bound on observer time)**.** *For comparable events $e_1 \prec e_2$ and any observer $O = (\leq_O, \mu_O)$:*

$$t_O(e_2) - t_O(e_1) \geq d_{\text{chain}}(e_1, e_2) \cdot \min_{e \in \mathcal{E}} \mu_O(e) \tag{7}$$

Proof. Let $e_1 = c_0 \prec c_1 \prec \cdots \prec c_k = e_2$ be a shortest chain ($k = d_{\text{chain}}(e_1, e_2)$). Since $\leq_O$ extends $\prec$, the observer ordering satisfies $c_0 \leq_O c_1 \leq_O \cdots \leq_O c_k$. Therefore all k events $c_1, \ldots, c_k$ lie in the interval $\{e' : e_1 <_O e' \leq_O e_2\}$. Each contributes at least $\min_e \mu_O(e)$ to the sum. □

Remark 19. *This proposition connects observer-dependent time to observer-independent structure: the chain-length separation provides a lower bound on any observer's elapsed time, scaled by the observer's minimum sampling rate. The structural interval "lives underneath" all observer times.*

8 Worked Example: Dual Pentagons

Example 20 (Pentagon separation structure)**.** *Using the dual-pentagon event set from Papers XXIII–XXIV: 10 events in two layers.*

Layer 0 *($\theta_b = 0$): events $(0,0), (1,1), (2,2), (3,3), (4,4)$.*

Layer 1 *($\theta_b = 2\pi/5$): events $(0,4), (1,0), (2,1), (3,2), (4,3)$.*

Partial order: *Every Layer-0 event precedes every Layer-1 event. Events within the same layer are incomparable.*

Cover relations: *Every Layer-0 event covers every Layer-1 event (since there are no intermediate events). So $e \lessdot f$ for all e in Layer 0 and f in Layer 1.*

Hasse graph: *Complete bipartite graph $K_{5,5}$ between the two layers.*

Chain-length separation (comparable pairs)

For any Layer-0 event e and Layer-1 event f: $d_{\text{chain}}(e, f) = 1$ (one cover step). All comparable pairs have the same chain-length separation.

Observer-disagreement separation (incomparable pairs)

Within Layer 0, events $(0, 0)$ and $(1, 1)$ are incomparable. Using two observers from Paper XXIV:

- *O_1: assigns $t_{O_1}(0, 0) = 1$, $t_{O_1}(1, 1) = 2$. Difference: -1.*
- *O_2: assigns $t_{O_2}(0, 0) = 10$, $t_{O_2}(1, 1) = 6$. Difference: $+4$.*

Mean difference: $(-1+4)/2 = 1.5$. Variance: $\sigma = ((11.5)^2+(41.5)^2)/2 = (6.25+6.25)/2 = 6.25$.

So $\sigma((0, 0), (1, 1)) = 6.25 > 0$: the incomparable events are distinguished by observer disagreement.

Transition-cost metric (all pairs)

The Hasse graph is $K_{5,5}$.

- *Comparable pair (different layers): $d_{\text{trans}} = 1$ (direct edge).*
- *Incomparable pair (same layer): $d_{\text{trans}} = 2$ (go to any event in the other layer, then back). For example, $(0, 0) \to (0, 4) \to (1, 1)$: path length 2.*

Summary

Table 2: Separations in the pentagon toy model.

Pair type	d_{chain}	σ	d_{trans}
Same event	0	0	0
Comparable (cross-layer)	1	> 0 (fixed sign)	1
Incomparable (same layer)	undefined	6.25 (sign-varying)	2

The transition-cost metric gives $d_{\text{trans}} = 1$ for "timelike" (comparable) pairs and $d_{\text{trans}} = 2$ for "spacelike" (incomparable) pairs. Incomparable events are farther in the transition metric than comparable ones — the sideways cost exceeds the forward cost.

9 Relation to Known Frameworks

Remark 21. *In causal set theory, the longest chain between comparable events approximates Lorentzian proper time, and spatial distance is estimated from causal-diamond volumes. The VFD approach differs in that the event set and order are derived from 600-cell geometry rather than postulated. The metric constructions here are structural parallels, not claimed equivalences.*

Table 3: Comparison with causal-set and graph-metric approaches.

Feature	VFD (this paper)	Causal set / graph theory
Temporal separation	Min-chain length	Longest chain (proper time)
Spatial precursor	Observer disagreement	Volume-based estimators
Unified metric	Hasse graph distance	Graph metric on Hasse diagram
Metric derivation	From event-order geometry	From assumed causal structure
Signed interval	Not yet constructed	Lorentzian distance (assumed)

10 Discussion and Limitations

10.1 What is established

- Cover relation and Hasse graph explicitly defined (Definitions 1–2).
- Chain-length separation: well-defined directed temporal separation on comparable pairs (Proposition 6).
- Observer-sign structure: comparable pairs have observer-independent sign; incomparable pairs have observer-dependent sign (Proposition 10). Variance σ provides a quantitative separation measure but does not satisfy the triangle inequality (explicitly noted).
- Transition-cost metric: a genuine metric on the full event set (Theorem 14).
- Structural bound relating observer time to chain length (Proposition 18).
- All constructions demonstrated on the pentagon toy model.

10.2 What is not established

- No Lorentzian metric. The transition-cost metric is positive-definite, not Lorentzian.
- No signed interval. A $ds^2 = dt^2 - dx^2$ decomposition is not derived.
- No continuous manifold. All constructions are discrete (graph-based).
- No Einstein equations or curvature. The metric is flat in the toy model.
- The observer-disagreement quantity is not a metric (triangle inequality failure is explicit).
- The Hasse graph connectivity (required for Theorem 14) is assumed, not proved for the full 600-cell event structure.

10.3 Metric failure modes (honest assessment)

- d_{chain}: only defined for comparable pairs; it is a directed temporal separation, not a metric on full $\mathcal{E}$.
- σ: fails triangle inequality; not a metric in the formal sense.

- d_{trans}: satisfies all axioms but is positive-definite, losing the timelike/spacelike distinction that Lorentzian geometry requires.

A future construction that recovers a signed interval from the interplay of d_{chain} and d_{trans} would represent a significant step toward Lorentzian structure. The present paper provides the ingredients.

11 Conclusion

The event-order framework of Papers XXIII–XXIV naturally supports emergent separation structure. Three constructions are developed:

- **Chain-length separation**: an observer-independent directed temporal separation for comparable events (properties proved).
- **Observer-disagreement separation**: a nontrivial observer-dependent separation measure for incomparable events, interpreted as a spatial precursor (sign structure proved; not a metric).
- **Transition-cost metric**: a genuine metric on the full event set from Hasse-graph shortest paths (metric axioms proved, conditional on graph connectivity).

The structural distinction between comparable (chain-based, temporal) and incomparable (sideways, spatial-precursor) separation mirrors the timelike/spacelike distinction of Lorentzian geometry, though a signed interval is not yet derived.

The event-order framework now carries: temporal ordering (Paper XXIII), observer-dependent kinematics (Paper XXIV), and separation geometry (this paper). What remains for the GR programme is the construction of curvature and dynamical equations on this structure.

References

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